

Lipids in Detail

LDL and ApoB (APOE, LDLR, PCSK9, APOB, HMGCR, SORT1, ABCG5/8), HDL and triglycerides (CETP, LIPC, LPL, APOA5, GCKR, TRIB1), Lp(a), response to statins and scores of all lipid fractions

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. It has **APOE ε3/ε3** and alleles that lower **LDL** in **LDLR** and **SORT1 (1p13)**, without APOB T71I or ABCG5/8 risk. The genomic scores of LDL are below the median (20.5 and 24.0 percentiles), and the total cholesterol in 16.6. Non-HDL 33.3, ApoB 46.1, hypercholesterolemia 38.9. Triglycerides have important protective genes (APOA5 –1131 and S19W without risk, LPL N291S absent). Lp(a) does not have any of the two very high value alleles (rs10455872, rs3798220). The genetic tolerance to statins is good (SLCO1B1 *1/*14 probable: increased function, without the allele of myopathy rs4149056; ABCG2 G/G).

Unfavourable. It has **PCSK9 rs11206510 T/T** (2 alleles of LDL higher), **APOB rs693** and **HMGCR** heterozygotes, and the score of 223 variants of LDL is at percentile 69.2. Triglycerides are genetically above the median (68.2 and 72.7 percentiles; loci load in 89.7), with GCKR, TRIB1 and ANGPTL3 heterozygotes. The Lp(a) scores are typical-high (61.1 and 75.0 percentiles). HDL and ApoA1 are low (18.2, 35.1 and 19.7 percentiles). This is without judgment (HDL is not causal), but follows the triglycerides.

What does it mean? Genetic LDL and ApoB are typical to favourable, without common family hypercholesterolemia or APOE ε4. This does not change the target: LDL/ApoB should be as low as possible. The weak point is the triglyceride-remanescent axis (TG ↑, HDL ↓), sensitive to refined hydrates, hepatic fat and alcohol (zero). Lp(a) is probably moderate; genes do not exclude it and the measured value orders (§26). If one day is necessary, statins are genetically well tolerated, with a typical response (percentile 61.8).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 26 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 25 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 4 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 15 PGS Catalog scores calculated locally with 9.328.209 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVABLE/ALTO; ≤ 20 : MENOS PROVABLE/BAIXO; between 20 and 80: TIPIC.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 12687 loci, 10965 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

High percentile : higher genetic value. For LDL, ApoB, non-HDL, total cholesterol, triglycerides and Lp(a), higher is worse. HDL and ApoA1 are left without judgment (high HDL is not a causal protector). In response to statins, higher = more descent of LDL.

LDL and ApoB

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
LDL	PGS000892 (Graham 2021, PRS-CS, 1.12 million variants)	1.085.899 1.119.211 (weighted coverage 96%)	20,5	TIPIC	typical
LDL	PGS002703 (Weissbrod 2022, SbayesR, 970 thousand variants)	948.324 970.081 (weighted coverage 97%)	24,0	TIPIC	typical
LDL	PGS000115 (Trinder 2020, 223 variants)	219 223 (weighted coverage 85%)	69,2	TIPIC	typical
ApoB	PGS002102 (Privé 2022, LDpred2, 315 thousand variants)	310.652 315.483 (weighted coverage 98%)	46,1	TIPIC	typical
Non-HDL cholesterol	PGS002782 (Kanoni 2022, PRS-CS, 1.24 million variants)	1.082.280 1.239.184 (weighted coverage 82%)	33,3	TIPIC	typical
Total cholesterol	PGS002718 (Weissbrod 2022, SbayesR, 977 thousand variants)	955.164 977.049 (weighted coverage 98%)	16,6	BAIXO	favourable
Hypercholesterolemia	PGS002700 (Weissbrod 2022, SbayesR, 938 thousand variants)	916.425 937.601 (weighted coverage 97%)	38,9	TIPIC	typical

HDL and triglycerides

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
HDL	PGS002695 (Weissbrod 2022, SbayesR, 972 thousand variants)	949.906 971.682 (weighted coverage 97%)	18,2	BAIXO	without judgment
HDL	PGS002172 (Privé 2022, LDpred2, 733 thousand variants)	720.453 732.902 (weighted coverage 98%)	35,1	TIPIC	without judgment
ApoA1	PGS002101 (Privé 2022, LDpred2, 681 thousand variants)	669.933 681.234 (weighted coverage 98%)	19,7	BAIXO	without judgment
Triglycerides	PGS002719 (Weissbrod 2022, SbayesR, 981 thousand variants)	959.485 981.472 (weighted coverage 97%)	68,2	TIPIC	typical
Triglycerides	PGS002197 (Privé 2022, LDpred2, 731 thousand variants)	718.601 731.035 (weighted coverage 98%)	72,7	TIPIC	typical

- **HDL:** High genetic HDL does not reduce cardiovascular risk (mendelian randomization studies); therefore, it has no judgment.

Lipoprotein(a)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Lp(a)	PGS000667 (Trinder 2020, 43 variants)	28 43 (weighted coverage 47%)	61,1	TIPIC	typical
Lp(a)	PGS000752 (Dron 2021, LDpred, 11.4 thousand variants)	10.808 11.446 (weighted coverage 90%)	75,0	TIPIC	typical

Response to statins

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
LDL descent with statin	PGS002730 (Mayerhofer 2022, 35 variants)	32 35 (weighted coverage 92%)	61,8	TIPIC	typical

- **LDL descent with statin:** High percentile = higher descent of LDL with statin.

4. Key variants by topic

Variants with replicated effect on blood lipids and pharmacogenetics of statins.

LDL and ApoB

LDL and ApoB are the direct cause of atherosclerosis. The lower the life, the better. Genes regulate hepatic uptake (LDLR, PCSK9, SORT1), synthesis (HMGCR, APOB) and intestinal cholesterol absorption (ABCG5/8).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
APOE	rs429358	ε4	T/T	imputation DR2=0.99	C (0 copy/s)	favourable	Defines ε4 (C allele): LDL ↑ ~10 mg/dL and more cholesterol absorption	Strong
APOE	rs7412	ε2	C/C	chip	T (0 copy/s)	neutral	Defines ε2 (T allele): LDL ↓ (but TG ↑ in ε2/ε2)	Strong
LDLR	rs6511720		G/T	chip	T (1 copy/s)	favourable	T allele increases LDL receptor expression (LDL ↓ ~7 mg/dL per allele)	Strong
PCSK9	rs11591147	R46L	G/G	chip	T (0 copy/s)	neutral	R46L: loss of PCSK9 function (LDL ↓ ~15%, ↓ ~30–45% infarction); rare	Strong
PCSK9	rs11206510		T/T	chip	T (2 copy/s)	unfavourable	Common locus of PCSK9: the T allele rises to LDL	Moderate
APOB	rs1367117	T71I	G/G	chip	A (0 copy/s)	favourable	T71I: allele A rises to LDL and ApoB	Moderate
APOB	rs693		G/A	chip	A (1 copy/s)	unfavourable (light)	Synonymous variant of APOB (XbaI): Allele A is associated with higher LDL	Moderate
CELSR2/ SORT1 (1p13)	rs12740374		G/T	chip	T (1 copy/s)	favourable	The T allele creates a C/EBP site that increases the SORT1 in the liver (LDL ↓ ~5 mg/dL per allele)	Strong
HMGCR	rs12916		T/C	chip	C (1 copy/s)	unfavourable (light)	Target of statins: the C allele rises to LDL (small effect)	Moderate
ABCG5/8	rs6544713		C/C	chip	T (0 copy/s)	favourable	Sterol transporters in the intestine and liver: T allele rises to LDL (more cholesterol absorption and phytosterols)	Moderate
ABCG8	rs11887534	D19H	G/G	chip	C (0 copy/s)	neutral	D19H: C allele increases biliary cholesterol excretion (LDL ↓ mild, but more stones in the vesicle)	Moderate

HDL and triglycerides

Triglycerides depend on lipolysis of triglyceride-rich lipoproteins (LPL, APOA5, ANGPTL3) and liver production (GCKR, TRIB1). The high HDL does not protect (CETP, LIPC): it loses judgment.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CETP	rs3764261		C/C	chip	A (0 copy/s)	without judgment	Allele A reduces CETP: HDL ↑ (high genetic HDL does not protect the heart)	Strong
CETP	rs708272	TaqIB	G/G	chip	A (0 copy/s)	without judgment	TaqIB (B2 = allele A): minus CETP, HDL ↑	Moderate
LIPC	rs1800588		C/C	chip	T (0 copy/s)	without judgment	Hepatic lipase –514C>T: the T allele lowers hepatic lipase (HDL ↑; Smaller LDL/mass)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LPL	rs328	S447X	C/C	chip	G (0 copy/s)	neutral	S447X: LPL function gain (TG ↓ ~10%, HDL ↑; less coronary disease)	Strong
LPL	rs268	N291S	A/A	chip	G (0 copy/s)	favourable	N291S: partial loss of LPL function (TG ↑); rare	Moderate
APOA5	rs662799		A/A	chip	G (0 copy/s)	favourable	–1131T>C: the G allele (C in gene nomenclature) rises to ~20% triglycerides	Strong
APOA5	rs3135506	S19W	G/G	chip	C (0 copy/s)	favourable	S19W: C allele (Trp19) reduces ApoA5 secretion (TG ↑)	Moderate
GCKR	rs1260326	P446L	T/C	chip	T (1 copy/s)	unfavourable (light)	P446L: the T allele rises to triglycerides (liver lipogenesis)	Strong
TRIB1	rs2954029		A/T	chip	A (1 copy/s)	unfavourable (light)	Allele A rises triglycerides and LDL	Moderate
ANGPTL3 (DOCK7)	rs2131925		G/T	chip	T (1 copy/s)	unfavourable (light)	ANGPTL3 locus: the T allele rises triglycerides	Moderate

Lipoprotein(a)

Lp(a) is almost all genetic (LPA). The two highest effect SNPs and the score say if a high value is likely. The real value is only known by measuring (see §26).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LPA	rs10455872		A/A	chip	G (0 copy/s)	favourable	Short kringle IV alleles Very high Lp(a) (OR ~1.5–1.7 for coronary heart disease)	Strong
LPA	rs3798220	I4399M	T/T	chip	C (0 copy/s)	favourable	I4399M: high Lp(a)	Strong
LPA	rs41272114		C/C	chip	T (0 copy/s)	neutral	Splicing variant with loss of function: Low Lp(a)	Moderate

Statins and other lipids

Pharmacogenetics: SLCO1B1 and ABCG2 regulate the levels of statin in blood and the risk of myopathy.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SLCO1B1	rs4149056	*5	T/T	chip	C (0 copy/s)	favourable	*5 (V174A): less hepatic uptake of statins (symvastatin, atorvastatin) and more myopathy	Strong
ABCG2	rs2231142	Q141K	G/G	chip	T (0 copy/s)	favourable	Q141K: less efflux of rosuvastatin (≈ levels; more effect and more side effects)	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each lipid, with alleles oriented towards higher value, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
LDL	2101 (1788)	1932 de 3576	1778	67,1	typical
Total cholesterol	2731 (2363)	2536 de 4726	2352	71,7	typical
ApoB	969 (861)	854 de 1722	855	21,5	typical
HDL	2815 (2474)	2456 de 4948	2464	16,9	without judgment
Triglycerides	3644 (3192)	3234 de 6384	3183	89,7	unfavourable
Lp(a)	427 (287)	291 de 574	286	13,7	favourable

6. Favourable

- **APOE ε3/ε3**: without ε4 (LDL ↑, Alzheimer ↑) and without ε2 (dysbetalipoproteinemia). **Strong.**
- **Polygenic to favourable LDL/ApoB**: LDL in the 20.5 and 24.0 percentiles (genomic), total cholesterol in 16.6, non-HDL in 33.3 and ApoB in 46.1. Loci load of ApoB in 21.5; LDL (67.1) and total cholesterol (71.7) are typical. **Moderate.**
- **LDLR rs6511720 G/T** and **SORT1 rs12740374 G/T**: 1 allele that lowers the LDL in each (SORT1 increases the hepatic uptake of LDL). **Moderate.**
- **APOB T71I G/G** and **ABCG5/8 C/C**: without alleles of LDL ↑. **Moderate.**
- **APOA5 –1131 A/A** and **S19W G/G**; **LPL N291S absent**: without the main variants of high triglycerides. **Strong.**
- **LPA rs10455872 A/A** and **rs3798220 T/T**: without 2 alleles of Lp(a) very high; loci load of Lp(a) at percentile 13.7; PGS of Lp(a) of 11446 variants gives p75 (typical) and KIV-2 is not visible in the chip, so only the analysis of Lp(a) confirms. **Moderate-strong.**
- **Well tolerated statins**: Probable SLCO1B1 *1/*14 (augmented function without rs4149056 myopathy allele: T/T; less myopathy with simvastatin and atorvastatin) and ABCG2 Q141K absent (G/G; normal levels of rosuvastatin). **Strong.**

7. Unfavourable

- **PCSK9 rs11206510 T/T**: 2 alleles of slightly higher LDL (+~2 mg/dL per allele). **Moderate.**
- **APOB rs693 G/A** and **HMGCR rs12916 T/C**: 1 allele LDL ↑ in each. **Weak moderated.**
- **LDL, 223 variant score**: percentile 69.2. It disagrees with the two genomics (which are below the median); genomics are the most reliable. **Weak.**
- **Triglycerides**: percentiles 68.2 and 72.7 (loci load in 89.7); GCKR P446L, TRIB1 and ANGPTL3 heterozygotes. **Moderate.**
- **Lp(a)**: typical-high scores (61.1 and 75.0 percentiles; the loci load, on the contrary, is low). Without the alleles of great effect, but the number of repetitions kringle IV-2 is not readable in the chip. **Weak moderated.**
- **Low HDL/ApoA1** (18.2, 35.1 and 19.7 percentiles; HDL loci load in 16.9): without causal judgment, but mark the standard TG ↑ / HDL ↓. **Information.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Vascular Aging	LDL/ApoB genetics typical to favourable; triglycerides and remnants slightly ↑ ; Lp(a) probably moderate	Mixed
Metabolic Homeostasis Decline	Triglycerides-HDL unfavourable axis (GCKR, TRIB1, ANGPTL3) against favourable APOA5/LPL	Neutral to unfavourable (mild)
Chronic Inflammation	Lp(a) and the remnants are pro-inflammatory in the arterial wall	Neutral
Oxidative stress	The oxidized LDL depends on the total LDL (here typical to low)	Neutral to favourable
Arterial stiffness	No high genetic LDL (APOE ε3/ε3, without common family variants)	Favourable (light)
Insulin resistance	GCKR binds glucose to hepatic lipogenesis and triglycerides (see the Glucose report)	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **LDL/ApoB as low as possible, regardless of favourable genetics.** The cumulative dose throughout life is what counts. *Suggestion (not applied).*
2. **High fiber, mainly soluble** (oat/beta-glucan, legumes, psyllium; the more, the better). Low LDL 5–10% and triglycerides. *Suggestion (not applied).*
3. **Low saturated fat, substituted by unsaturated fat** (rice, dried fruits). Phytosterols 2 g/day are an option (ABCG5/8 without hyperabsorption). *Suggestion (not applied).*
4. **Triglycerides:** free sugars and refined flours at least; Omega-3 3.2–4 g/day (already in the plane; low TG 20–30%); zero alcohol (already comply); regular aerobic exercise. *Suggestion (not applied).*
5. **Measure Lp(a) once** if it is not yet measured (and ApoB annually). Genetics does not exclude it (see §26). *Suggestion (not applied).*
6. **If the doctor indicates hypolipemia:** genetically well tolerated statin (SLCO1B1 *1/*14 probable, with increased function and without the allele of myopathy rs4149056; Normal ABCG2) and with typical response. Ezetimiba and inhibitors of PCSK9 are possible complements. Medical decision. *Suggestion (not applied).*
7. **Lipid panel with ApoB and fasting triglycerides, annual.** The remnants (total cholesterol – HDL – LDL) are useful with this TG profile. *Suggestion (not applied).*

10. Limitations

- The chip does not read the rare mutations of familial hypercholesterolemia (LDLR, APOB, PCSK9 rare), so a very high measured LDL would continue to require sequencing.
- The number of kringle repetitions IV-2 of LPA (the main determinant of Lp(a)) is not measured with the chip. The Lp(a) scores capture only part of the variation.
- The three LDL scores disagree (223 variants vs genomics); the genomics are the most complete.
- HDL is presented without judgment: genetically high HDL does not reduce cardiovascular risk.
- The load of loci GWAS sums SNPs in linkage imbalance. It is descriptive.
- Conclusions only from genes; it is not a diagnosis. The measured values are in the §23 and §26 reports.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** 15 published scores (LDL, ApoB, non-HDL, total cholesterol, hypercholesterolemia, HDL, ApoA1, triglycerides, Lp(a) and statins response). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Graham 2021 and Willer 2013 for lipids); Musunuru 2010 for SORT1; Clarke 2009 for LPA; SEARCH 2008 and CPIC 2022 for SLCO1B1 and statins), GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
LDL and ApoB	APOE	rs429358	T/T	imputation DR2=0.99	C	0	strand +	without the risk allele
LDL and ApoB	APOE	rs7412	C/C	chip	T	0	strand +	without the protective allele (common genotype)
LDL and ApoB	LDLR	rs6511720	G/T	chip	T	1	strand +	1 copy of the protective allele
LDL and ApoB	PCSK9	rs11591147	G/G	chip	T	0	strand +	without the protective allele (common genotype)
LDL and ApoB	PCSK9	rs11206510	T/T	chip	T	2	strand +	2 risk allele copies
LDL and ApoB	APOB	rs1367117	G/G	chip	A	0	strand +	without the risk allele
LDL and ApoB	APOB	rs693	G/A	chip	A	1	strand +	1 copy of risk allele
LDL and ApoB	CELSR2/ SORT1 (1p13)	rs12740374	G/T	chip	T	1	strand +	1 copy of the protective allele
LDL and ApoB	HMGCR	rs12916	T/C	chip	C	1	strand +	1 copy of risk allele
LDL and ApoB	ABCG5/8	rs6544713	C/C	chip	T	0	strand +	without the risk allele
LDL and ApoB	ABCG8	rs11887534	G/G	chip	C	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
HDL and triglycerides	CETP	rs3764261	C/C	chip	A	0	strand +	0 copy of the effect allele
HDL and triglycerides	CETP	rs708272	G/G	chip	A	0	strand +	0 copy of the effect allele
HDL and triglycerides	LIPC	rs1800588	C/C	chip	T	0	strand +	0 copy of the effect allele
HDL and triglycerides	LPL	rs328	C/C	chip	G	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
HDL and triglycerides	LPL	rs268	A/A	chip	G	0	strand +	without the risk allele
HDL and triglycerides	APOA5	rs662799	A/A	chip	G	0	strand +	without the risk allele
HDL and triglycerides	APOA5	rs3135506	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
HDL and triglycerides	GCKR	rs1260326	T/C	chip	T	1	strand +	1 copy of risk allele
HDL and triglycerides	TRIB1	rs2954029	A/T	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
HDL and triglycerides	ANGPTL3 (DOCK7)	rs2131925	G/T	chip	T	1	strand +	1 copy of risk allele
Lipoprotein(a)	LPA	rs10455872	A/A	chip	G	0	strand +	without the risk allele
Lipoprotein(a)	LPA	rs3798220	T/T	chip	C	0	strand +	without the risk allele
Lipoprotein(a)	LPA	rs41272114	C/C	chip	T	0	strand +	without the protective allele (common genotype)
Statins and other lipids	SLCO1B1	rs4149056	T/T	chip	C	0	strand +	without the risk allele
Statins and other lipids	ABCG2	rs2231142	G/G	chip	T	0	strand +	without the risk allele